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MALWARE AND ANTIVIRUS

Mata Kuliah Dasar Keamanan Siber

Tri Febrianto, CEI.,CEH.,CHFI.,EHE

CASE STUDY

Pengguna Klik BCA Terserang 'Malware Pencuri Uang'

Trisno Heriyanto | CNN Indonesia

Kamis, 05 Mar 2015 18:12 WIB

Bagikan :

Jakarta, CNN Indonesia -- Sejumlah pengguna layanan internet banking KlikBCA telah menjadi korban pencurian uang. Belasan juta raib dalam sekejap akibat program jahat komputer atau populer disebut malware.

"Kemungkinan komputer korban terkena malware Zeus," kata Alfons, dari vaksincom saat dihubungi CNN Indonesia.

https://www.cnnindonesia.com/teknologi/20150305181253-185-37063/pengguna-klik-bca-terserang-malware-pencuri-uang

CASE STUDY

Dua Rumah Sakit di Jakarta Kena Serangan Ransomware WannaCry

Lesthia Kertopati | CNN Indonesia

Sabtu, 13 May 2017 19:40 WIB

Bagikan :

Ilustrasi: RS Dharmais dan RS Harapan Kita di Jakarta ikut terdampak serangan ransomware WannaCry yang juga melanda dunia. (Thinkstock/Korawig)

Jakarta, CNN Indonesia-- Kementrian Komunikasi dan Informatika Republik Indonesia (Kominfo) menyebut dua rumah sakit di Jakarta menjadi korban serangan siber ransomware WannaCry, yang juga melanda dunia.

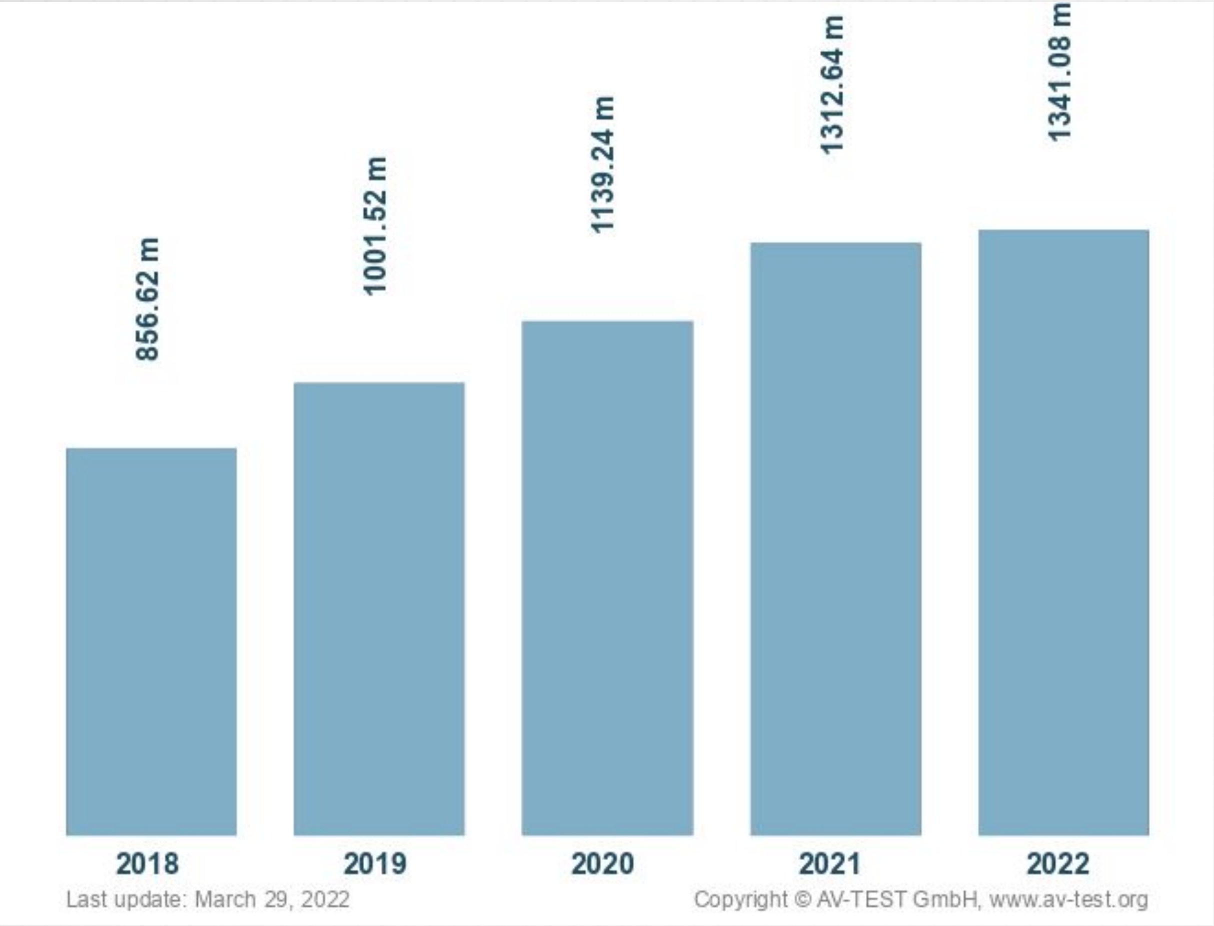
Dua rumah sakit yang dimaksud adalah Rumah Sakit Harapan Kita dan Rumah Sakit Dharmais.

https://www.cnnindonesia.com/teknologi/20150305181253-185-37063/pengguna-klik-bca-terserang-malware-pencuri-uang

MODULE OBJECTIVE

- Understand what Malware is
- Identify the various types of Malwares
- Identify the various symptoms of Malware infection
- Understand what an antivirus is and how it works
- Understand how to choose the right antivirus software for you and what its limitations are
- Configure and use Kaspersky and Avast antivirus software
- Test if your antivirus is working

TOTAL MALWARE FOUND



Lima besar yang menyerang Indonesia

1. Trojan-Ransom.Win32.Wanna
2. Trojan-Ransom.Win32.Stop
3. Trojan-Ransom.Win32.Cryakl
4. Trojan-Ransom.Win32.GandCrypt
5. Trojan-Ransom.Win32.Gen

https://www.av-test.org/en/statistics/malware/

WHAT IS MALWARE

- Malware, short for "malicious software," refers to any intrusive software developed by cybercriminals (often called "hackers") to steal data and damage or destroy computers and computer systems.

7 TYPE OF MALWARE

1. **Viruses** are a subgroup of malware. A virus is malicious software attached to a document or file that supports macros to execute its code and spread from host to host. Once downloaded, the virus will lay dormant until the file is opened and in use. Viruses are designed to disrupt a system’s ability to operate. As a result, viruses can cause significant operational issues and data loss.
2. **Worms** are a malicious software that rapidly replicates and spreads to any device within the network. Unlike viruses, worms do not need host programs to disseminate. A worm infects a device via a downloaded file or a network connection before it multiplies and disperses at an exponential rate. Like viruses, worms can severely disrupt the operations of a device and cause data loss
3. **Trojan** viruses are disguised as helpful software programs. But once the user downloads it, the Trojan virus can gain access to sensitive data and then modify, block, or delete the data. This can be extremely harmful to the performance of the device. Unlike normal viruses and worms, Trojan viruses are not designed to self-replicate.
4. **Spyware** is malicious software that runs secretly on a computer and reports back to a remote user. Rather than simply disrupting a device’s operations, spyware targets sensitive information and can grant remote access to predators. Spyware is often used to steal financial or personal information. A specific type of spyware is a keylogger, which records your keystrokes to reveal passwords and personal information.

7 TYPE OF MALWARE

5. **Adware** is malicious software used to collect data on your computer usage and provide appropriate advertisements to you. While adware is not always dangerous, in some cases adware can cause issues for your system. Adware can redirect your browser to unsafe sites, and it can even contain Trojan horses and spyware. Additionally, significant levels of adware can slow down your system noticeably. Because not all adware is malicious, it is important to have protection that constantly and intelligently scans these programs.
6. **Ransomware** is malicious software that gains access to sensitive information within a system, encrypts that information so that the user cannot access it, and then demands a financial payout for the data to be released. Ransomware is commonly part of a phishing scam. By clicking a disguised link, the user downloads the ransomware. The attacker proceeds to encrypt specific information that can only be opened by a mathematical key they know. When the attacker receives payment, the data is unlocked.
7. **Fileless malware** is a type of memory-resident malware. As the term suggests, it is malware that operates from a victim's computer's memory, not from files on the hard drive. Because there are no files to scan, it is harder to detect than traditional malware. It also makes forensics more difficult because the malware disappears when the victim computer is rebooted

SYMPTOMS OF MALWARE

- Slow System
- Unexpected Crashes
- Explicit or Excessive Pop Ups
- Excessive and Suspicious Hard Drive Activity
- Antivirus and Firewall Disabling
- New Browser Home Page or Toolbar
- Blacklisted IP Address
- Peculiar or Uninitiated Program Activity
- Random Network Activity

WINDOWS ARCHITECTURE AND OPERATIONS WINDOWS FILE SYSTEMS

NTFS formatting creates important structures on the disk for file storage, and tables for recording the locations of files:

- **Partition Boot Sector:** This is the first 16 sectors of the drive. It contains the location of the Master File Table (MFT). The last 16 sectors contain a copy of the boot sector.
- **Master File Table (MFT):** This table contains the locations of all the files and directories on the partition, including file attributes such as security information and timestamps.
- **System Files:** These are hidden files that store information about other volumes and file attributes.
- **File Area:** The main area of the partition where files and directories are stored.

Note: *When formatting a partition, the previous data may still be recoverable because not all the data is completely removed. It is recommended to perform a secure wipe on a drive that is being reused. The secure wipe will write data to the entire drive multiple times to ensure there is no remaining data.*

ALTERNATE DATA STREAMS

- NTFS stores files as a series of attributes, such as the name of the file, or a timestamp.
- The data which the file contains is stored in the attribute $DATA, and is known as a data stream.
- By using NTFS, Alternate Data Streams (ADSs) can be connected to the file.
- An attacker could store malicious code within an ADS that can then be called from a different file.
- In the NTFS file system, a file with an ADS is identified after the filename and a colon, for example, **Testfile.txt:ADS**. This filename indicates an ADS called ADS is associated with the file called **Testfile.txt**.

```
C:\ADS> echo "Alternate Data Here" > Testfile.txt:ADS
C:\ADS> dir
 Volume in drive C is Windows
 Volume Serial Number is A606-CB1B
 Directory of C:\ADS
2020-04-28  04:01 PM    <DIR>          .
2020-04-28  04:01 PM    <DIR>          ..
2020-04-28  04:01 PM                 0 Testfile.txt
               1 File(s)              0 bytes
               2 Dir(s)  43,509,571,584 bytes free
C:\ADS> more < Testfile.txt:ADS
"Alternate Data Here"
C:\ADS> dir /r
 Volume in drive C is Windows
 Volume Serial Number is A606-CB1B
 Directory of C:\ADS
2020-04-28  04:01 PM    <DIR>          .
2020-04-28  04:01 PM    <DIR>          ..
2020-04-28  04:01 PM                 0 Testfile.txt
                                    24 Testfile.txt:ADS:$DATA
               1 File(s)              0 bytes
               2 Dir(s)  43,509,624,832 bytes free
C:\ADS>
```

SANDBOXING KOTAK PASIR

- Sandboxing is a technique that allows suspicious files to be executed and analyzed in a safe environment.
- Cuckoo Sandbox is a popular free malware analysis system sandbox. It can be run locally and have malware samples submitted to it for analysis.
- ANY.RUN is an online tool that offers the ability to upload a malware sample for analysis like any online sandbox.

ANTI VIRUS

- Antivirus is a software application that detects and eradicates malware from a computer system. Some popular antivirus software are McAfee, Norton, AVG, and Avast.
- Antivirus software employs the following techniques to keep computers secure:
  1. Scanning - Scanning is a virus detection technique which checks all the files on a system and looks for malicious code. Its is of two types—on access and on demand.
  2. Integrity Checking - This is a virus detection technique that compares the current state of data to a previously known good state and checks for discrepancies.

HOW DOES ANTIVIRUS TAKE ACTION AGAINST INFECTED FILES?

- When an antivirus program comes across an infected file, three actions can be taken - clean, quarantine, or delete
- Quarantine is an antivirus software function which isolates files suspected of being infected with malware. These files are isolated to prevent them from further corrupting other files in the system. It's similar to quarantining an individual who has contracted something contagious and possibly dangerous

HOW TO CHOOSE THE RIGHT ANTIVIRUS SOFTWARE?

- Compatibility - An antivirus software can only be effective on a system if its compatible with other applications on the system
- Usability - Antivirus software should be easy to understand for the users
- Comprehensive Protection - Antivirus software must offer protection around the clock. It should also be able to detect malware in a system Quality of Protection :
  - Effectiveness of malware detection processes
  - Frequency and regularity of updates
  - Ability to remove infections from computer systems
  - Efficiency in delivering comp

HOW TO CHOOSE THE RIGHT ANTIVIRUS SOFTWARE?

TEST YOUR ANTI VIRUS

- https://en.wikipedia.org/wiki/EICAR_test_file

```
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
```

TEST YOUR ANTI VIRUS

Kerjakan Tugas Lab I